

WEEK 6 FINAL PROJECT REPORT

Student Management System

Student Name	Harshini P V
Internship Domain	Python
Internship	InternNova Python Internship
Week	6 of 6
Project Title	Student Management System
Technology	Python 3
Version Control	Git & GitHub

Submitted as part of the Week 6 internship requirements.

1. Introduction

The Student Management System is a menu-driven Python application developed as the final project for Week 6 of the InternNova Python Internship. The project demonstrates Python fundamentals, data structures, file handling, exception handling, standard libraries, and Object-Oriented Programming. Git and GitHub are used for version control and project management.

2. Problem Statement

Managing student information manually can make it difficult to add, search, update, display, and store records consistently. The project provides a simple console-based solution for managing student records.

3. Objectives

- Develop a practical Python application using concepts learned during the internship.
- Create and manage student records through a menu-driven interface.
- Apply classes, objects, inheritance, polymorphism, encapsulation, and abstraction.
- Use dictionaries and file handling for record management.
- Handle invalid user input using exception handling.
- Use the datetime library to display the current date and time.
- Use Git and GitHub to maintain and publish the project.

4. Project Features

- Add Student – accepts name, age, course, and marks.
- View Students – displays all stored records and calculated grades.
- Search Student – searches using Student ID.
- Update Marks – changes marks with 0–100 validation.
- Delete Student – removes a selected student record.
- Save Data to File – stores student information in students.txt.
- Show Date and Time – displays the current date and time.
- Demonstrate Polymorphism – demonstrates Student and Teacher role behavior.

5. Technologies Used

Technology / Tool	Purpose
Python 3	Application development
Visual Studio Code	Writing and running Python code
Git	Local version control
GitHub	Remote repository and project sharing
Python standard library	abc and datetime functionality

6. Python Concepts Implemented

Variables and Data Types: Names, ages, courses, marks and IDs use suitable Python data types.

Input and Output: input() collects user data and print() displays menus and results.

Type Casting: Input is converted to int or float for numeric operations.

Conditional Statements: if/elif/else handles validation, menu choices and grades.

Loops: A while loop keeps the application running until Exit.

Functions: Separate methods perform add, view, search, update, delete and save operations.

Dictionaries: Student records are maintained using Student ID as the key.

File Handling: Records are written to students.txt.

Exception Handling: try/except handles invalid input and file errors.

Libraries: datetime displays the current date and time; abc supports abstraction.

7. Object-Oriented Programming Concepts

Class and Object: Person, Student, Teacher and StudentManagementSystem are classes.

Constructor: __init__() initializes object attributes.

Inheritance: Student and Teacher inherit common behavior from Person.

Abstraction: Person is an abstract base class with display_role().

Encapsulation: Private __marks is accessed through getter/setter methods.

Polymorphism: Student and Teacher provide different display_role() implementations.

8. Project Workflow

1. Start the program
2. Display the menu
3. User selects an operation
4. Validate input
5. Create/manage Student objects
6. View/search/update/delete records
7. Save data to students.txt
8. Continue until Exit

9. Git and GitHub Implementation

Git was initialized in the Student_Management_System project folder. The project files were staged and committed, the main branch was used, GitHub was connected as the remote origin, and the project was successfully pushed to GitHub. README.md and the Screenshots folder were also uploaded.

Repository: InternNova_Python-Internship

10. Screenshots

Screenshots of Git/GitHub activity and the Student Management System execution are included in the submitted project materials.

11. Challenges Faced

- Working in the correct Git project directory.
- Configuring Git with the GitHub no-reply email after the original commit email was rejected.
- Organizing the repository with README.md and Screenshots.
- Handling invalid user input safely.
- Implementing multiple OOP concepts in one application.
- Capturing clear terminal output screenshots.

12. Future Improvements

- Add a graphical user interface (GUI).
- Use SQLite or another database.
- Add login and authentication.
- Add sorting and advanced search.
- Generate student performance reports.
- Export records to CSV or PDF.

13. Conclusion

The Student Management System successfully demonstrates the Python concepts required for the Week 6 final project. It combines Python fundamentals, data structures, file handling, exception handling, standard libraries and OOP in a practical application. Git and GitHub were also used to manage and publish the project.

14. Student / Internship Details

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